

NAKHONSI THAMMARAT AGROMET, Thailand

WMO: 485540

Lat: 8.359N

Lon: 100.059E

Elev: 10

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.9	73.1	69.4	108.3	76.4	70.5	112.4	77.5	17.2	83.0	14.2	81.2	0.1	N/A	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.6	95.4	81.3	94.2	81.1	93.1	80.9	82.8	91.8	82.4	91.1	81.8	90.2	6.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.7	159.3	87.7	80.1	156.0	86.8	79.7	154.0	86.3	46.9	92.0	46.2	91.5	45.6	90.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	11.9	10.4	DB	69.2	97.1	2.4	1.7	67.5	98.4	66.2	99.4	64.8	100.3	63.1	101.6	(4)
(5)			WB	67.9	84.0	2.1	0.7	66.4	84.5	65.2	84.9	64.0	85.3	62.5	85.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	79.8	80.4	82.1	84.1	84.8	84.5	84.1	84.1	83.3	82.1	80.6	80.0
	DBStd	2.56	1.77	1.43	1.77	2.39	1.67	1.75	1.78	1.61	1.72	1.87	2.07	1.83
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11863	924	852	997	1024	1078	1034	1056	1057	998	995	919	929
	CDD65	6388	459	432	532	574	613	584	591	592	548	530	469	464
	CDH74	64491	3900	3996	5533	6473	6751	6520	6376	6259	5448	5023	4225	3987
	CDH80	23833	986	1341	2189	2813	2835	2756	2632	2599	2040	1648	1099	895
Wind	WSAvg	2.4	2.2	1.9	1.8	1.7	2.0	2.8	3.4	3.2	2.7	2.0	2.1	2.8
Precipitation	PrecAvg	64.4	2.9	1.1	2.4	3.0	5.0	5.4	5.7	5.7	5.7	8.5	11.3	7.5
	PrecMax	101.6	17.6	6.8	16.7	7.8	8.0	7.6	8.5	9.1	8.3	15.1	27.7	18.2
	PrecMin	41.3	0.1	0.1	0.1	0.2	3.0	1.9	2.4	3.3	3.5	4.0	5.0	2.0
	PrecStd	11.2	3.3	1.4	3.1	1.9	1.2	1.3	1.6	1.5	1.2	2.5	5.3	4.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.9	90.0	94.0	96.9	96.4	95.7	95.8	96.2	94.3	92.3	90.3	89.2
		MCWB	79.4	78.8	81.4	82.3	82.1	81.1	81.0	79.9	80.6	81.0	80.2	80.2
	2%	DB	87.5	88.8	91.8	95.0	94.4	94.1	94.0	94.2	92.5	90.5	88.7	87.6
		MCWB	79.1	78.5	80.2	81.9	81.8	80.9	80.7	80.3	80.5	80.6	80.4	79.7
	5%	DB	86.4	87.8	90.4	93.1	93.0	92.7	92.5	92.6	91.2	89.2	87.4	86.3
		MCWB	78.5	78.3	79.7	81.2	81.7	80.8	80.3	80.2	80.4	80.5	80.2	79.3
	10%	DB	85.2	86.8	89.1	91.4	91.3	91.2	90.7	90.7	89.5	87.9	85.9	84.8
		MCWB	78.0	77.9	79.3	80.7	81.3	80.6	80.0	80.0	80.0	80.2	80.0	78.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	80.7	82.5	83.9	83.7	83.3	82.3	82.5	82.5	82.4	81.9	81.6
		MCDB	86.3	87.6	91.5	94.2	92.7	92.7	92.1	91.9	90.4	89.4	87.3	86.3
	2%	WB	80.0	79.8	81.2	82.7	82.8	82.3	81.6	81.7	81.7	81.6	81.4	80.7
		MCDB	85.4	86.6	89.7	92.4	91.7	91.6	91.1	90.7	89.7	88.3	86.6	85.8
	5%	WB	79.3	79.1	80.4	81.9	82.3	81.6	80.9	80.9	81.0	80.9	80.7	80.0
		MCDB	84.4	85.8	88.4	90.8	90.9	90.4	90.1	89.8	89.1	87.5	85.9	84.7
	10%	WB	78.7	78.5	79.8	81.1	81.6	80.8	80.3	80.3	80.3	80.4	80.1	79.3
		MCDB	83.4	84.9	87.1	89.3	89.7	89.3	88.9	88.8	88.1	86.6	84.7	83.5
Mean Daily Temperature Range	5% DB	MDBR	10.8	14.0	14.6	14.6	14.6	14.5	14.6	15.2	13.9	12.2	9.5	9.1
		MCDBR	12.8	15.1	16.7	17.4	16.7	16.6	16.9	17.5	16.2	14.4	12.5	11.9
		MCWBR	5.2	6.1	6.4	5.9	5.9	5.5	5.5	5.7	5.7	5.7	5.4	4.9
	5% WB	MCDBR	11.3	14.1	15.5	16.2	15.6	15.4	15.5	15.8	15.1	13.4	11.4	10.5
		MCWBR	5.3	6.2	6.3	6.1	6.1	5.9	5.6	5.9	6.0	5.9	5.5	5.0
Clear-Sky Solar Irradiance	taub	0.495	0.499	0.507	0.478	0.456	0.453	0.455	0.449	0.442	0.454	0.476	0.486	
	taud	2.168	2.148	2.163	2.252	2.352	2.386	2.368	2.373	2.386	2.358	2.274	2.215	
	Ebn at Noon	260	263	262	266	267	264	265	270	276	273	264	260	
	Edh at Noon	48	50	50	45	40	38	39	39	39	40	43	45	
All-Sky Solar Radiation	RadAvg	1466	1774	1839	1830	1640	1573	1553	1599	1565	1405	1251	1246	
	RadStd	137	116	155	110	92	82	92	74	62	114	121	132	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	+0.38	+0.45	N/A	+0.56	+0.38	+0.43	N/A	N/A	+133	+135

Nomenclature: See separate page